

647. Palindromic Substrings

647. Palindromic Substrings

Medium

 Topics

 Companies

 Hint

Given a string `s`, return *the number of **palindromic substrings** in it.*

A string is a **palindrome** when it reads the same backward as forward.

A **substring** is a contiguous sequence of characters within the string.

Example 1:

Input: `s = "abc"`

Output: 3

Explanation: Three palindromic strings: "a", "b", "c".

Example 2:

Input: `s = "aaa"`

Output: 6

Explanation: Six palindromic strings: "a", "a", "a", "aa", "aa", "aaa".

First you understand the question:

- **no of palindromic substring.**

note: Palindrome can be two types:

- **odd length palindrome.**

e.g: **abcba**

- **even length palindrome.**

e.g: **abccba**

we have to check the both possibilities.

Brute force approach:-

- Iterate Through Each Character.
- Double Loop.
- Check Palindrome.
- Count Palindromes.
- Update Overall Count
- Repeat for All Characters.
- Final Result.

C++   Auto

```
1  class Solution {
2  public:
3      bool isPalindrome(string &s, int l, int r){
4          while(l <= r){
5              if(s[l] != s[r]) return false;
6              l++, r--;
7          }
8          return true;
9      }
10     int countSubstrings(string s) {
11         int n = s.size();
12         int ans = 0;
13         for(int i=0; i<n; i++){
14             int cnt = 0;
15             for(int j=i; j<n; j++){
16                 if(isPalindrome(s, i, j)) cnt++;
17             }
18             ans += cnt;
19         }
20         return ans;
21     }
22 };
```

- **Time complexity: $O(N^3)$**
- **Space complexity: $O(1)$**

Accepted Runtime: 0 ms

• Case 1 • Case 2

Input

```
s =  
"abc"
```

Output

```
3
```

Expected

```
3
```



follow me on:



<https://www.linkedin.com/in/manish-kumar-689905213/>



manishkumar_978